

DIY PROJECT IDEAS

FOR YOUR NEW SCHOOL YEAR

WANT DIY IDEAS TO WOW YOUR TEACHERS AND PEERS AND JUST TO HAVE SOME FUN THIS YEAR? BEHOLD!

JAYESH SHINDE

jayesh@digit.in

Vacation's over. You're back in school. Time to stick to the curriculum, execute all your assignments on time, and leave no stone unturned in your quest to get good grades come examination time. Is this all that there is to school life? Well, here's a little something we've put together to help you earn some brownie points and enhance your reputation on campus as a DIY nerd.

1 LAUNCH A PHONE INTO SPACE

At first this idea may sound absurd, but think about it – if you have an old smartphone lying around, you can attach it to a high-altitude weather bal-



loon and launch it as high as 30 to 50 km above the earth's surface. Straight into the stratosphere!

Purchase the balloon (with a burst altitude of 100,000 feet) for as low as ₹2,000 to ₹3,000 (like this one <http://goo.gl/DeM2gR>), go to a recreational space, find a balloon seller, pay him to fill your weather balloon with Helium, attach your old smartphone to it with video recording turned on, attach a small powerbank to it for good measure (make sure the whole payload is under 500g), launch your balloon. Also, read about NASA's PhoneSat initiative and ways to better protect your smartphone in extreme cold and hot weather (<http://goo.gl/zTL78M>), if you're seriously planning this stunt. For a slightly more serious minded project, check this out: <http://goo.gl/28HVc3>.

2 MAKING A PCB

Whether you're a student of electronics or not, it doesn't matter. The fact that a printed circuit board (or PCB) is inside of pretty much all our digital toys is reason enough to get your hands dirty and try making the fun-

damental building block yourself. Consider it as a rite of passage. With just tools available in your home or garage, you can make your own PCB – yes, without a CNC machine or the rest of the jazz.



Just get a copper clad board, some hydrogen peroxide (the stuff ENT specialists use to clean your ears with), a hand drill machine and a laser printer. Oh, and don't forget the schematics of an actual PCB for you to be able to test! For detailed steps, go here: <http://goo.gl/UIN345>. Piece of cake, right?

3 BUILD YOUR OWN DRONE

Don't worry if you can't afford the latest and greatest quadcopters launched in the market – DJI Phantom,